

Fabio Aurelio D'Asaro

Assistant Professor in Logic (PHIL-02/A)

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Researcher in Logic (PHIL-02/A) working at the interface of formal epistemology and AI: probabilistic and temporal reasoning, argumentation, and logic programming (ASP/ILASP). Teaching and supervision span Logic and Philosophy of Science, AI, theoretical and applied Computer Science, Epistemology of Big Data, and Logic Programming.

EXPERIENCE

Assistant Professor, University of Verona 2026 – present
Verona, Italy

Research and teaching on uncertain and epistemic reasoning, temporal and probabilistic logics, logic programming, reasoning about actions and change, bounded reasoning, and formal argumentation for (explainable) AI.

Honorary Lecturer, University College London (UCL) 2020 – present
London, UK

Department of Information Studies, Faculty of Arts and Humanities. Continuing collaboration on explainable AI and the Event Calculus (KIDS group). Email: fabio.dasaro.14@ucl.ac.uk — profiles.ucl.ac.uk/46877-fabio-d'asaro

Postdoctoral Researcher, University of Salento 2025 – 2026
Lecce, Italy

Department of Human Studies (PHIL-02/A). Project FAIR — Future AI Research (PE00000013), NRRP/MUR, NextGenerationEU, Spoke 6 ("Symbiotic AI"), PI: Paolo Baldi.

Assistant Professor, University of Verona 2022 – 2025
Verona, Italy

Research and teaching on uncertain and epistemic reasoning, temporal and probabilistic logics, logic programming, reasoning about actions and change, bounded reasoning, and formal argumentation for (explainable) AI. Partially funded by PON R&I 2014-2020, DM 1062/2021 (Action IV.4); UniVR code 2021rtdaPON03; CUP B39J21025840001; 12-month secondment at REVO.

Postdoctoral Researcher, Logic Group, University of Milan 2020 – 2021
Milan, Italy

Uncertain, depth-bounded and epistemic logics.

Postdoctoral Researcher, CRDC Tecnologie 2019 – 2020
Naples, Italy

AVATEA Project (AI to support rehabilitation of children with neuro-motor disorders); explainable AI for robotics and preference learning.

Research Assistant, University of Palermo 2013 – 2014
Palermo, Italy

NEVERLOST Project (PO-FESR 2007-2013): (sub)optimal antenna displacement in arbitrary 2D spaces via soft-computing (fuzzy logic, genetic algorithms); educational "serious game" for theoretical CS.

Undergraduate Intern, ICAR-CNR Palermo Jun – Sep 2011
Palermo, Italy

IMPULSO project: a path-visualization tool using the Google Maps APIs.

EDUCATION

PhD in Artificial Intelligence 2014 – 2019
University College London (UCL), UK

Thesis: "Probabilistic Epistemic Reasoning About Actions" (supervisors: Rob Miller, Antonis Bikakis, Luke Dickens). Modelling uncertainty and knowledge-producing actions in the Event Calculus. Excerpts published in Artificial Intelligence (AIJ) and LPNMR.

MSc in Pure Mathematics and Mathematical Logic

2013 – 2014

University of Manchester, UK

Distinction ($\approx 110/110$ cum laude). Dissertation: "Analogical Reasoning in Unary Inductive Logic" (supervisor: Jeff B. Paris).

BSc in Computer Science

2006 – 2011

University of Palermo, Italy

110/110 cum laude. Dissertation on the Church-Turing thesis and new models of computation (supervisor: Settimo Termini).

PUBLICATIONS (Citations 281, h -index 10, $i10$ 13)

2026

- M. Fanghella, C. F. Colombo, **F. A. D'Asaro**, M. T. Pascarelli, G. Barchiesi, M. Rabuffetti, M. Ferrarin, F. Guala and C. Sinigaglia. *Rational expectations and kinematic information in coordination games*. Cognition 271. doi:10.1016/j.cognition.2026.106453
- M. Fanghella, **F. A. D'Asaro**, D. Quarona, G. Barchiesi, M. Rabuffetti, M. Ferrarin and C. Sinigaglia. *Early Target Object Prediction in Action Observation*. Quarterly Journal of Experimental Psychology. doi:10.1177/17470218261442536
- D. Fossemò, F. Mignosi, G. Placidi, L. Raggioli, M. Spezialetti and **F. A. D'Asaro**. *Explaining Neural Networks in Preference Learning: a Post-hoc Inductive Logic Programming Approach*. (arXiv preprint)

2025

- A. Aquino, **F. A. D'Asaro**, R. Gaudenzi, M. Lezcano, V. Iacovella and M. Vezzoli. *AI in the Experimental Loop: Implications for Replicability in Social Robotics and Social Sciences*. (Preprint)
- A. Bikakis, **F. A. D'Asaro**, A. Diallo, L. Dickens, A. Hunter and R. Miller. *A Graphical Formalism for Reasoning about Substitution in Resource Transforming Procedures*. Journal of Artificial Intelligence Research. doi:10.1613/jair.1.18606
- **F. A. D'Asaro**, F. Genco and G. Primiero. *Checking trustworthiness of probabilistic computations in a typed natural deduction system*. Journal of Logic and Computation. doi:10.1093/logcom/exaf003
- P. Baldi and **F. A. D'Asaro**. *Non-monotonic Bounded Reasoners*. The Reasoner 19(1). doi:10.54103/1757-0522/27548
- P. Baldi, **F. A. D'Asaro**, A. Dyoub and F. A. Lisi. *Weighted Assumption Based Argumentation to reason about ethical principles and actions*. Proceedings of CILC 2025.
- L. Xu, L. Dickens and **F. A. D'Asaro**. *A Translation of Probabilistic Event Calculus into Markov Decision Processes*. Proceedings of TIME 2025. (Accepted, in press)

2024

- R. Gaudenzi, M. Scandola, **F. A. D'Asaro**, V. Moro and M. Badino. *"Sensing" the future: From Kant to the brain as a predictor*. doi:10.22541/au.171945748.89099414/v1 (Preprint (ESS Open Archive / Authorea))
- A. Aquino, **F. A. D'Asaro**, M. Lezcano, V. Iacovella, M. Scandola and M. Vezzoli. *How can we conceive replicability with AI-mediated experimental conditions? A position talk*. Atti del Congresso AIP Sperimentale 2024.
- M. Badino, **F. A. D'Asaro** and F. Pedrazzoli. *Educare all'IA. La sfida didattica dell'Intelligenza Artificiale: ChatGPT e Gemini*. Sanoma Italia. (ISBN 9791256030019)
- **F. A. D'Asaro**, A. Bikakis, L. Dickens and R. Miller. *An Answer Set Programming-based implementation of Epistemic Probabilistic Event Calculus*. International Journal of Approximate Reasoning. doi:10.1016/j.ijar.2023.109101

2023

- L. Raggioli, **F. A. D'Asaro** and S. Rossi. *Deep Reinforcement Learning for Robotic Approaching Behavior Influenced by User Activity and Disengagement*. International Journal of Social Robotics. doi:10.1007/s12369-023-01044-7

- **F. A. D'Asaro**, L. Raggioli, S. Malek, M. Grazioso and S. Rossi. *An Application of a Runtime Epistemic Probabilistic Event Calculus to Decision-making in e-Health Systems*. Theory and Practice of Logic Programming. doi:10.1017/S1471068422000382
- Z. Mileva, A. Bikakis, **F. A. D'Asaro**, M. Law and A. Russo. *A Unifying Framework for Learning Argumentation Semantics*. arXiv preprint arXiv:2310.12309.
- G. Coraglia, **F. A. D'Asaro**, F. Genco, D. Giannuzzi, D. Posillipo, G. Primiero and C. Quaggio. *BRIOrAlkemy: A Bias Detecting Tool*. Proceedings of BEWARE-23 (AIXIA 2023).
- M. D'Agostino, **F. A. D'Asaro** and C. Larese. *Advancing the Boundaries of Formal Argumentation: Reflections on the AI3 Special Issue*. Journal of Applied Logics.
- F. Pedrazzoli, **F. A. D'Asaro** and M. Badino. *How do Decision Support Systems Nudge?*. Proceedings of CEPE 2023.

2022

- D. Fossemò, F. Mignosi, L. Raggioli, M. Spezialetti and **F. A. D'Asaro**. *Using Inductive Logic Programming to globally approximate Neural Networks for preference learning: challenges and preliminary results*. Proceedings of BEWARE-22 (AIXIA 2022).
- G. Primiero and **F. A. D'Asaro**. *Proof-checking bias in labeling methods*. Proceedings of BEWARE-22 (AIXIA 2022).
- B. S. Caffo, **F. A. D'Asaro**, A. d'Avila Garcez and E. Raffinetti. *Explainable artificial intelligence models and methods in finance and healthcare*. Frontiers in Artificial Intelligence. doi:10.3389/frai.2022.970246

2021

- A. Termine, G. Primiero and **F. A. D'Asaro**. *Modelling Accuracy and Trustworthiness of Explaining Agents*. Proceedings of LORI 2021. doi:10.1007/978-3-030-88708-7_19
- **F. A. D'Asaro**, P. Baldi and G. Primiero. *Introducing k-lingo: a k-depth bounded version of ASP system clingo*. Proceedings of KR 2021. doi:10.24963/kr.2021/65
- **F. A. D'Asaro** and G. Primiero. *Probabilistic Typed Natural Deduction for Trustworthy Computations*. Proceedings of the 22nd Int. Workshop on Trust in Agent Societies (TRUST 2021 @ AAMAS).
- G. D. Acciaro, **F. A. D'Asaro** and S. Rossi. *Predicting humans: a sensor-based architecture for real time Intent Recognition using ProbLog*. Proceedings of WOA 2021.

2020

- **F. A. D'Asaro**, A. Bikakis, L. Dickens and R. Miller. *Probabilistic Reasoning About Epistemic Action Narratives*. Artificial Intelligence 287. doi:10.1016/j.artint.2020.103352
- S. Sangiovanni, M. Spezialetti, **F. A. D'Asaro**, G. Maggi and S. Rossi. *Administrating Cognitive Tests Through HRI: an Application of an Automatic Scoring System Through Visual Analysis*. Proceedings of ICSR 2020.
- **F. A. D'Asaro**, M. Spezialetti, L. Raggioli and S. Rossi. *Towards an Inductive Logic Programming approach for explaining black-box preference learning systems*. Proceedings of KR 2020. doi:10.24963/kr.2020/88

2019

- **F. A. D'Asaro**, A. Origlia and S. Rossi. *Towards a Logic-Based Approach for Multi-Modal Fusion and Decision Making during Motor Rehabilitation Sessions*. Proceedings of WOA 2019.

2017

- **F. A. D'Asaro**, M. A. Di Gangi, V. Perticone and M. E. Tabacchi. *Computational Intelligence and Citizen Communication in the Smart City*. Informatik-Spektrum 40(1).
- **F. A. D'Asaro**, A. Bikakis, L. Dickens and R. Miller. *Foundations for a Probabilistic Event Calculus*. Proceedings of LPNMR 2017.

2016

- **F. A. D'Asaro** and J. B. Paris. *A Note on Carnap's Continuum and the Weak State Description Analogy Principle*. (Unpublished)

2015

- F. D'Aleo, **F. A. D'Asaro**, V. Perticone, G. Rizzo and M. E. Tabacchi. *Agents Displacement in Arbitrary Geometrical Spaces: An Evolutionary Computation based Approach*. Proceedings of ICAART 2015.

2014

- **F. A. D'Asaro**, V. Perticone and M. E. Tabacchi. *L'obiezione di una Lady ed il computer che vince ai telequiz*. flessIBILMENTE: un modello sistemico di approccio al tema della flessibilità, Pensa Multimedia.

2013

- **F. A. D'Asaro**, V. Perticone, M. E. Tabacchi and S. Termini. *Reflections on Technology and Human Sciences: rediscovering a common thread through the analysis of a few epistemological features of fuzziness*. Archives for the Philosophy and History of Soft Computing(1).
- **F. A. D'Asaro**, V. Perticone and M. E. Tabacchi. *A fuzzy methodology to alleviate information overload in eLearning*. Proceedings of EUSFLAT 2013.
- **F. A. D'Asaro**, V. Perticone, M. E. Tabacchi and S. Termini. *Technology and human sciences: a dialogue to be constructed or a common thread to be rediscovered?*. Proceedings of IFSA/NAFIPS 2013.

TEACHING

Lecturer, University of Verona

2022 – present

- Logic and Philosophy of Science (M-FIL/02), BA in Philosophy
- Epistemology of Big Data (M-FIL/02), MSc in Data Analytics
- Computational Epistemology (M-FIL/02), MSc in Artificial Intelligence
- Logic and Philosophy of Science in Physiotherapy (M-FIL/02), BSc
- Lab of Computer Science and Multimedia Technologies (INF/01), BA in Philosophy
- Lab of Artificial Intelligence and Neuroscience, PhD in Human Sciences

Lecturer, 8th Winter School in Linguistics (Verona / Free Univ. of Bolzano)

2025

- Introduction to Machine Learning (1.5h)

Lecturer, 3rd & 4th Summer School on Bayesian Statistical Analysis (Verona)

2023, 2025

- Introduction to Bayesian Probability (3h)

Lecturer, IECS Doctoral School, University of Trento

2023

- Logic Programming and Explainable AI (20h intensive, PhD)

Guest Lecturer, University of Milan (Dept. of Philosophy)

2021

- Topics in Logic, Probability and Computation (PhD; with H. Hosni, P. Baldi)
- Trends in Machine Learning (MSc, in H. Hosni's Probability Logic)

Teaching Assistant, University College London

2015 – 2023

- INST0074 Machine Reasoning for AI (module lead on Formal Argumentation, 2022/23)
- INST0060 Foundations of Machine Learning and Data Science
- INST0019 Introduction to Programming and Scripting; INST0029; INST0004

Teaching Assistant, University of Palermo

2013 – 2014

- Theoretical Computer Science (lab demonstrator)

INVITED TALKS

- Jun 2026 — Invited talk, PACMAN 2026 (Proof, Argumentation, Computation, Modalities and Negation), University of Verona
- May 2025 — Learning Argumentation Semantics, PACMAN 2025, Roma Tre University

- Mar 2024 — Logic programming for Explainable AI, PACMAN 2024, University of Verona
- Jun 2023 — Explainable AI applications of ILP, University of Salento
- Apr 2023 — Explainable AI applications of ILP, University of Lugano
- Mar 2022 — Probabilistic epistemic reasoning about actions, UCL, UK
- Dec 2021 — Explaining black-boxes using ILASP, University of Trieste
- Aug 2021 — "Probabilistic Reasoning About Epistemic Action Narratives", IJCAI 2021
- Apr 2021 — Invited demo, Lorentz Center Workshop on Explainable Medical AI
- Jun 2020 — Explaining black-boxes using ILASP, SPIKE Group, Imperial College London

PROJECTS

FAIR (PNRR)

2025 – present

Logic for ethical human-AI collaboration (eDefAI, "Symbiotic AI"). PE00000013, Spoke 6, CUP H97G22000210007.

REVO (PON)

2022 – 2024

Researcher (Verona): handwriting recognition, entity matching, eligibility pre-screening; 12-month secondment at REVO.

BRIO (PRIN)

2022 – 2025

Proponent / collaborator (Milan; then external, EThOS Verona). Formal frameworks for trust and bias. PRIN 2020, 2020SSKZ7R_001 (ERC SH4).

AVATEA (POR)

2018 – 2019

Runtime epistemic probabilistic Event Calculus for sensor fusion in rehabilitation serious games. POR Campania FESR 2014-2020, CUP B13D18000130007.

SUPERVISION & MENTORING

- Zlatina Mileva (Imperial College London) — Outstanding Project Award; preprint on learning argumentation semantics
- Daniele Fossemò & Marco D'Aviero (L'Aquila) — ILP for XAI; open dataset and BEWARE-22 paper
- Veronica Zenatelli (Verona) — BA thesis on LLMs and the Frame Problem; benchmark dataset
- Sara Sangiovanni (Napoli Federico II) — HRI pipeline for automated test administration; ICSR 2020
- Gennaro Daniele Acciaro (Napoli Federico II) — intent recognition with ProbLog/Event Calculus; WOA 2021
- Luca Raggioli (Napoli Federico II) — deep RL for human-aware approaching; IJSR 2023
- Francesco Pedrazzoli (Verona, PhD) — ethics of recommender systems; CEPE-23

PROFESSIONAL SERVICE

- Co-organizer, eDefAI — 3rd Logic for the AI Spring Summer School, Lecce, 2025
- Workshop Chair, 1st-3rd BEWARE workshops (Bias, Ethical AI, Explainability, Logic), AIxIA, since 2022
- Program Committee, IJCAI 2020, 2024, 2025
- Program Committee, KR, since 2022
- Program Committee, ECAI, since 2023
- Co-organizer, EThOS seminar series, since 2022
- Co-editor, Special Issue on Explainable AI in Finance and Healthcare, Frontiers in AI
- Workshop Chair, 1st ME&E (Machine Ethics and Explainability), ICLP 2021
- Workshop Chair, 5th AI³ (Advances in Argumentation in AI), AIxIA 2021
- Co-organizer, LUCI Lab seminars (Milan, Philosophy), 2020/2021
- Review Editor, editorial boards of Frontiers in AI, Frontiers in Big Data, Frontiers in AI in Neurology

REVIEWING

- Journals: IJAR (2025), Journal of Logic and Computation (NETREASON SI), AMAI (Commonsense SI), J. Hydrologic Engineering

- Conferences: KR (2022-2025), IJCAI (2016-2019, 2024, 2025), ECAI 2023, AAAI-18, ICLP-17, ICSR-2020, SMC-19, WOA-19

RESEARCH GROUPS

- EThOS — Ethics and Technologies Of the Self, University of Verona
- LUCI — Logic, Uncertainty, Computation and Information, University of Milan
- SPIKE — Imperial College London
- KIDS — Knowledge, Information and Data Science, UCL

MEMBERSHIPS

- AILA — Associazione Italiana di Logica e sue Applicazioni
- SILFS — Società Italiana di Logica e Filosofia delle Scienze
- AIxIA — Associazione Italiana per l'Intelligenza Artificiale
- GULP — Gruppo ricercatori e Utenti di Programmazione Logica

SKILLS

- **Logic programming:** ASP/clingo, Prolog/ProbLog, ILASP/FastLAS
- **Programming:** Python, Java, JavaScript/TypeScript, PHP, SQL, HTML/CSS; C, MATLAB; Lisp-family, Anglican

LANGUAGES

Italian — native • English — full professional proficiency

INTERESTS

Cinema · Guitar · Music · Chess